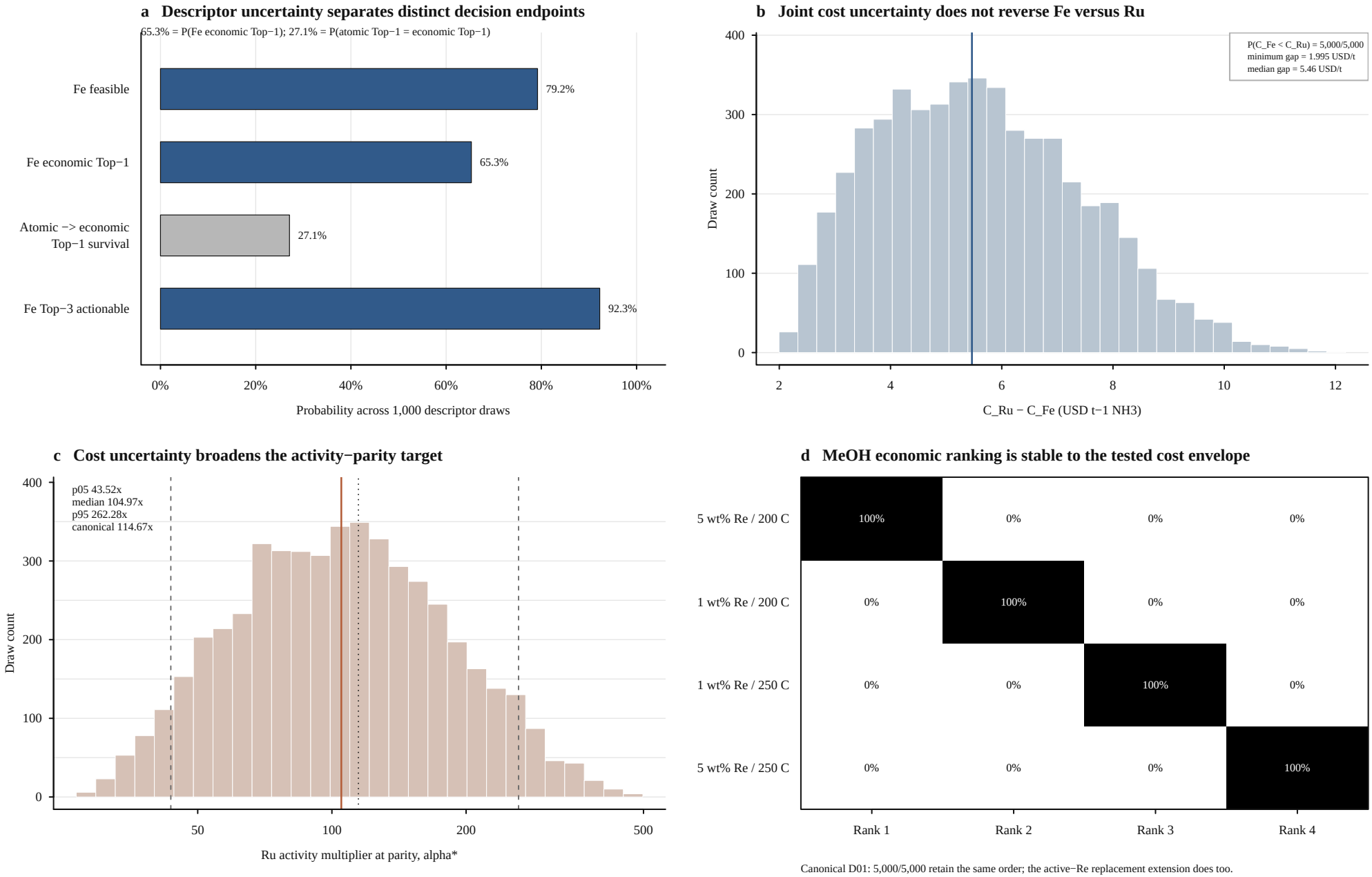
a Descriptor uncertainty separates distinct decision endpoints
65.3% = P(Fe economic Top−1); 27.1% = P(atomic Top−1 = economic Top−1)
Fe feasible
79.2%
Fe economic Top−1
65.3%
Atomic −> economic Top−1 survival
27.1%
Fe Top−3 actionable
92.3%
0%
20%
40%
60%
80%
100%
Probability across 1,000 descriptor draws
b Joint cost uncertainty does not reverse Fe versus Ru
P(C_Fe < C_Ru) = 5,000/5,000
minimum gap = 1.995 USD/t
median gap = 5.46 USD/t
Draw count
0
100
200
300
400
2
4
6
8
10
12
C_Ru − C_Fe (USD t−1 NH3)
c Cost uncertainty broadens the activity−parity target
p05 43.52x
median 104.97x
p95 262.28x
canonical 114.67x
Draw count
0
100
200
300
400
50
100
200
500
Ru activity multiplier at parity, alpha*
d MeOH economic ranking is stable to the tested cost envelope
5 wt% Re / 200 C
1 wt% Re / 200 C
1 wt% Re / 250 C
5 wt% Re / 250 C
100%
0%
Rank 1
Rank 2
Rank 3
Rank 4
Canonical D01: 5,000/5,000 retain the same order; the active−Re replacement extension does too.